

REQUIREMENTS ON SPEECH CODERS IMPOSED BY SPEECH SERVICE SOLUTIONS IN CELLULAR SYSTEMS

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ABSTRACT

Many digital cellular systems have been put into operation during the last few years in North America, Europe and Asia. The growth rate is high and the penetration is steadily increasing. New forecasts talk about 480 million subscribers in the world at the end of year 2000. The number of services and key features are increasing and will do so as the market is expanding. The trend is to build this on existing platforms, but beyond this an evolution into a generation 3 for wireless wideband, one called Universal Mobile Telecommunication Services (UMTS), has already started. In this paper we will point out different aspects of the requirements on speech coding algorithms that are seen in the cellular systems. It will be shown that to meet the diverse needs and applications no single speech and channel coding scheme is sufficient. We will see a need of flexibility and multiple speech coders that can meet the different requirements on quality, capacity and functionality.

1 INTRODUCTION

As the cellular industry is growing for each year and has reached a penetration above 25 % in some countries there is an increasing demand for enhancements of the voice service, the main communication mean. Today, standards as GSM, DAMPS, IS-95 and PDC [1] includes both first generation full-rate (FR) coders and second generation enhanced-FR (EFR) or half-rate (HR) coders. The second generation coders are intended to solve either the quality or the capacity issue for existing macro-cell systems.

The next generations of systems will contain both traditional macro-cell structures, new low cost micro cell systems, indoor business systems, private home systems (home base stations), advanced data-services and multimedia applications [2]. With many new users, private and business, with the growing business competition, and many advanced applications new requirements on the speech technology will be seen by the operators. All this implies that new speech coding schemes should solve a diverse set of requirements.

Many different standardization bodies address these items. Examples are ITU-R TG8/1 work on the FPLMTS program, ITU-T SG16 work on mobile use of G72x series, MPEG work on multimedia standards, ETSI SMG UMTS development and the cellular standardizations in ETSI, TTA and ARIB. Two main trends are seen. One is to continue the track on specific speech optimized services with full interaction between speech coding and the associated channel. Another is a more generic access service with little or no interaction between speech coders and channel protection.

2 REQUIREMENTS

To point out the different aspects of the requirements on future speech coding algorithms in cellular systems different speech service environments will be described.

The main environment today in a mobile telephone system is a cell structure with larger cells (0.5 - 40 km) usually called macro cellular or high-tier system. The speech coding used is referred to as FR. With a growing number of users and growing expectation the requirement for higher intrinsic quality is increasing and has resulted in standardization of EFR coders. It is exemplified by the requirement that background sounds should sound natural and be reproduced correctly. Enhanced quality (e.g. higher signal robustness) to reach true wireline quality will continuously be a possible improvement factor.

Another increasing demand is coverage. Good coverage is a main theme for customer satisfaction. Under some constraints one option is a more robust scheme, which should function in lower C/I situations than regular FR at the same gross bit-rate. This imposes a lower bit rate speech coder and is referred to as robust EFR (REFR). To achieve a perceived improvement the increased robustness need to be significant.

The next environment to look at is the high-density part of the cellular system. It is characterized as hot spots with both high and low mobility. Here the capacity is the main issue. The option is to reduce the speech coding bit-rate. Either highly protected speech coding allowing higher load or using half the gross bit-rate (HR) are conceivable. To enable higher quality of the HR service a low robustness extension for high C/I conditions could be allowed.

A third environment is a pico cell system for indoor coverage in shopping malls, business buildings or even homes. Very high quality is a main requirement to enable convergence of fixed and mobile services. One option is to increase speech coder bitrate assuming that lower robustness due to higher C/I, with less BER, is allowed. An increase of the gross bit-rate in low loaded indoor environments is also conceivable. Important is high quality background sound coding. In this environment true wireline quality, or even superior, in all aspects is required.

As cellular systems will be seen as just part of an integrated telecommunication system the use of multimedia services need to be enabled. This will require higher RF bandwidth to support audiovisual information. For the voice part we will see simultaneous voice and data and interaction between audio and video. Handsfree operation will be a requirement. The full benefit of using high quality wideband audio (7 kHz) will then be reached.

Evaluation of speech coders is a key issue for selecting solutions that can meet the defined requirements for a certain application. The traditional way to assess the subjective quality is to run a formal ACR test according to the ITU P.80-series. However, that methodology have several drawbacks even though it often provides very stable results. Results from conversational tests show that coders with close ACR results still differ substantially in the perceived speech quality. This fact indicates a great need to further develop

the methodology for speech quality testing to provide much more accurate and faithful results.

3 SPEECH AND CHANNEL CODING SOLUTIONS

3.1 Multi-Rate

A multi-rate, MR, coding concept is primarily focused on providing a flexible way to offer the best possible speech quality provided the current radio environment. The general concept should however be seen as a speech service toolbox possible to tailor to specific needs. The concept could be made more or less flexible based on each operators individual traffic planning, e.g. speech coding net bit rate and sufficient channel protection could be allocated according to the characteristics of the mode control. The different modes include the available gross channels (e.g. HR, FR, 2xFR etc.), and the supported partitionings of source and channel coding bit-rate.

The mode selection is then based on several control parameters; e.g. radio link transmission quality, system load, geographical position etc. The fast mode switching and quality estimation makes use of a fast (inband) control channel between the MS and the system. The time constant for mode switching could also be chosen to provide each operator with the freedom to optimize his system, i.e. mode selection is made at call set-up only, continuously during a call or even at system configuration (static selection).

Several MR design possibilities exist (e.g. use of existing coders, scalable coders, embedded coders etc.). The major challenge is however to design a scheme which allows efficient implementation (RAM/ROM, complexity, and system impact) and still offering high quality and service flexibility.

3.2 Variable-Rate

Source-controlled variable rate (VR) coding takes advantage of the strongly time-varying information flow in speech. VR coding may be used on its own, but can also provide increased flexibility and increased quality in a MR framework. The assumption that the input signal is produced by a composite-source model of stationary sources leads to multi-mode (MM) approaches to VR coding. The simplest form of a two-modal VR codec is widely realized as conventional fixed rate codec with discontinuous transmission (DTX). More advanced MM coders provide coding modes for different classes of the signal, e.g. for voiced, transient and unvoiced speech.

Studies show that systems employing VR codecs could potentially use less transmission capacity for the same quality level, than systems with fixed rate codecs, if robust classification is achieved. Comparisons of conventional fixed rate codecs with DTX with more advanced MM VR codecs show that most of the capacity gain, in both cases, is due to the distinction of speech activity and inactivity. However, MM VR codecs can lead to speech quality improvements at low capacity costs by including high-rate high-fidelity coding modes, provided that they occur relatively infrequent. The MM approach requires fast mode changes when ever there is a change of source signal class. Thus, seamless mode changes are of fundamental importance which makes it advantageous to accommodate scaled version of one common core coder in all coding modes.

3.3 Network - Service interaction

The integration of speech service (application) and the network (transport) in terms of unequal error protection (UEP) and use of additional link quality information can provide higher overall quality (medium to high error rates).

This integration does however not provide much flexibility. Addition of new services in this case affect several parts of the network. Conversely, a network which provides a generic transport to support 'packetized bit-streams' is fully flexible.

To fulfill both quality and flexibility aspects, the network may provide a range of pre-defined, protection classes. This may be requested and defined at connection setup. Due to changing radio conditions in cellular systems, high error rates can be expected. Satisfactory quality under errors can only be achieved if the application is informed about errors in the received data (on a frame or sub-frame basis). This information can be supplied either by error detection in the application (e.g. by CRC), or by the network. While the former provides for easier separation of application and transport the latter is preferred from a quality perspective.

It is proposed that new generic speech services are defined with an end-to-end protocol which allows the network to provide standardized link-quality information to the speech service. In this way, proper error concealment actions can be taken to ensure quality.

3.4 Error protection

Error protection for mobile speech services has in most existing systems been tailored to the specific coder, e.g. by use of UEP and error detection. When many different modes (bit-rates) are used, such as for MR and VR applications, it may be necessary to support a range of channel codecs with different rates. Different sub-streams of the data may also require different protection (UEP). To accommodate this, a set of channel coding rates must be provided. Using such generic FEC schemes, both network integrated and network independent applications can be realised.

Alternative approaches such as joint source and channel coding/decoding (e.g. by COVQ) to achieve robustness have been proposed. The higher overall quality can however only be achieved at the expense of higher transmission bandwidth in the network, i.e. the gross bit-rate must be transported and potentially also soft information.

Design of suitable protection schemes is a compromise between performance, transmission bandwidth and flexibility. The best performing scheme where all (soft) information is accumulated along the transmission and where a joint decoding is performed at the decoder requires very high bandwidth, but provides high flexibility. The current trends for speech services suggest that traditional FEC schemes are still the most viable.

4 SUMMARY

It has been shown that the cellular technology will exhibit a versatile environment in the future, which requires a set of speech and channel coding solutions. A main requirement will be flexibility in quality, robustness, bit-rate and functionality to meet the diverse environments and applications. An applicable solution is flexible network-controlled multi-mode coding with and without network interaction dependent on application. Source-controlled bit-rate could be a possible extension. A revised assessment technology that can distinguish between grades of naturalness as well as noisiness is needed to select coder solutions.

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